

TrES-3b

R-2817

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150801 · created 2026-09-05T02:05:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457235.8567 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.87 +/- 2.26 [%]
Orbital Inclination (inc)	85.06 +/- 2.87
Airmass coefficient 1 (a1)	0.35 +/- 0.2
Airmass coefficient 2 (a2)	0.54 +/- 0.19
Scatter in the residuals of the lightcurve fit is	49.64 %
Best Comparison Star	#1 - [521, 439]
Optimal Aperture	1.46
Optimal Annulus	7.25
Transit Duration (day)	0.072 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.197 $\pm$ 0.057 vs 0.16309 (deviation 0.41 σ)
Duration fit vs published	0.072 d vs 0.0567 d (deviation 0.59 σ)
Mid-time O-C	0.1 min
Depth SNR	2.4
Residual scatter	49.64 %

Light curve

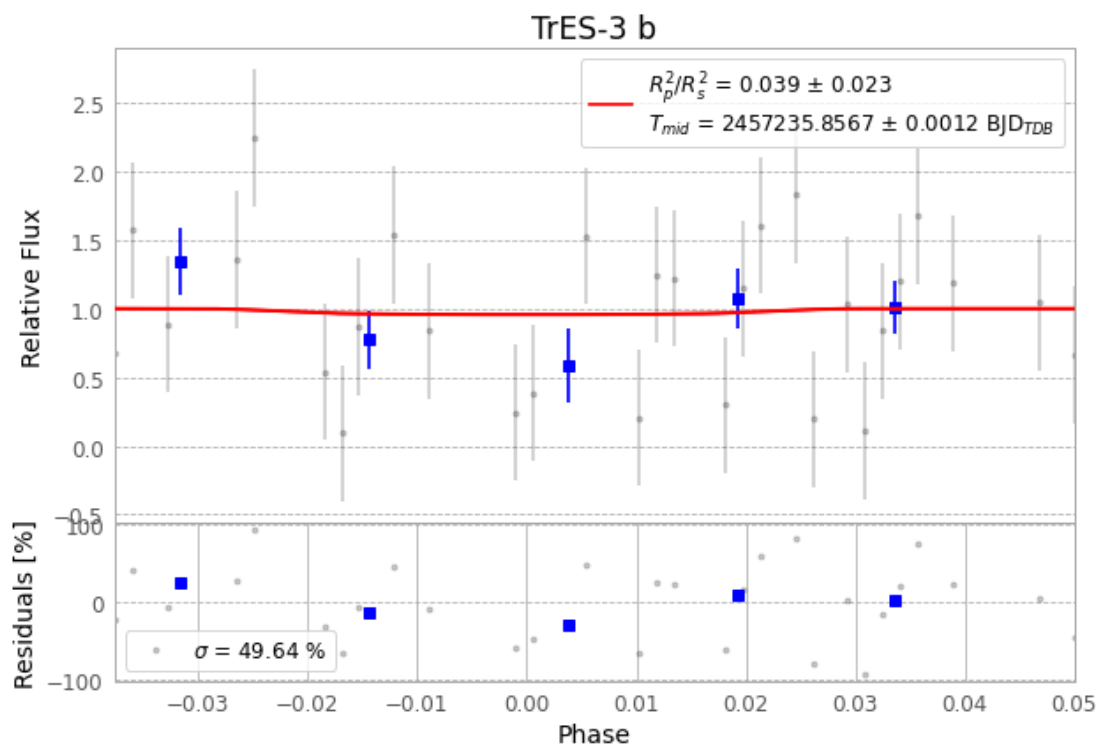


Figure 1 — FinalLightCurve_TrES-3 b_2015-07-31.png (SHA-256 7ad60c965b97b1ae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec4fc64583eba867...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3150801065712.FITS → TRES-3150801100912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150801.log (18ba23415cb7...), FinalLightCurve_TrES-3 b_2015-07-31.png (7ad60c965b97...), FinalLightCurve_TrES-3 b_2015-07-31.csv (fa9b2f362ca5...)

Compute resources

Wall time 304.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 02:05Z.